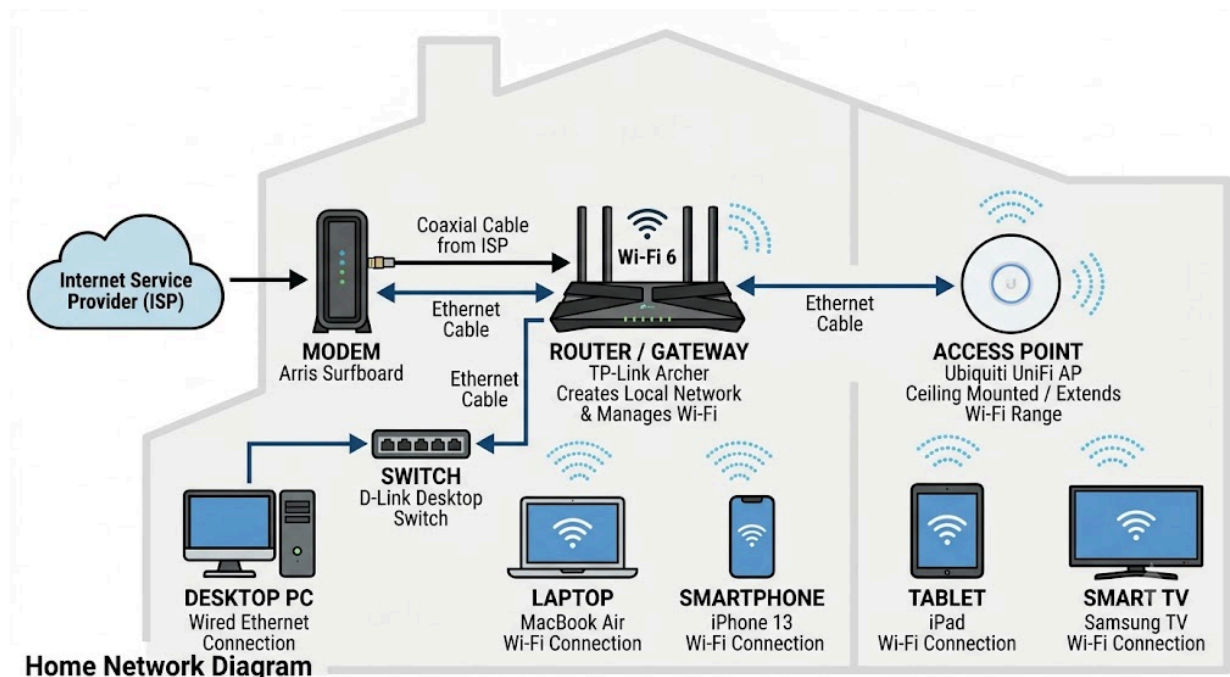


Q: Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

ANS:



Q: List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

ANS:

Modem: Translates ISP signals into internet.

- *Example:* In a home connected to a cable or fiber line.

Router: Directs internet traffic to multiple devices.

- *Example:* At home sharing Wi-Fi to phones and laptops.

Switch: Connects devices together on a wired local network.

- *Example:* In an office linking multiple desktop PCs.

Access Point: Broadcasts Wi-Fi to extend wireless coverage.

- *Example:* On a hotel ceiling providing guest Wi-Fi.

Network Cable: Physically carries data between devices.

- *Example:* Plugged into a gaming console for low latency.

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

ANS:

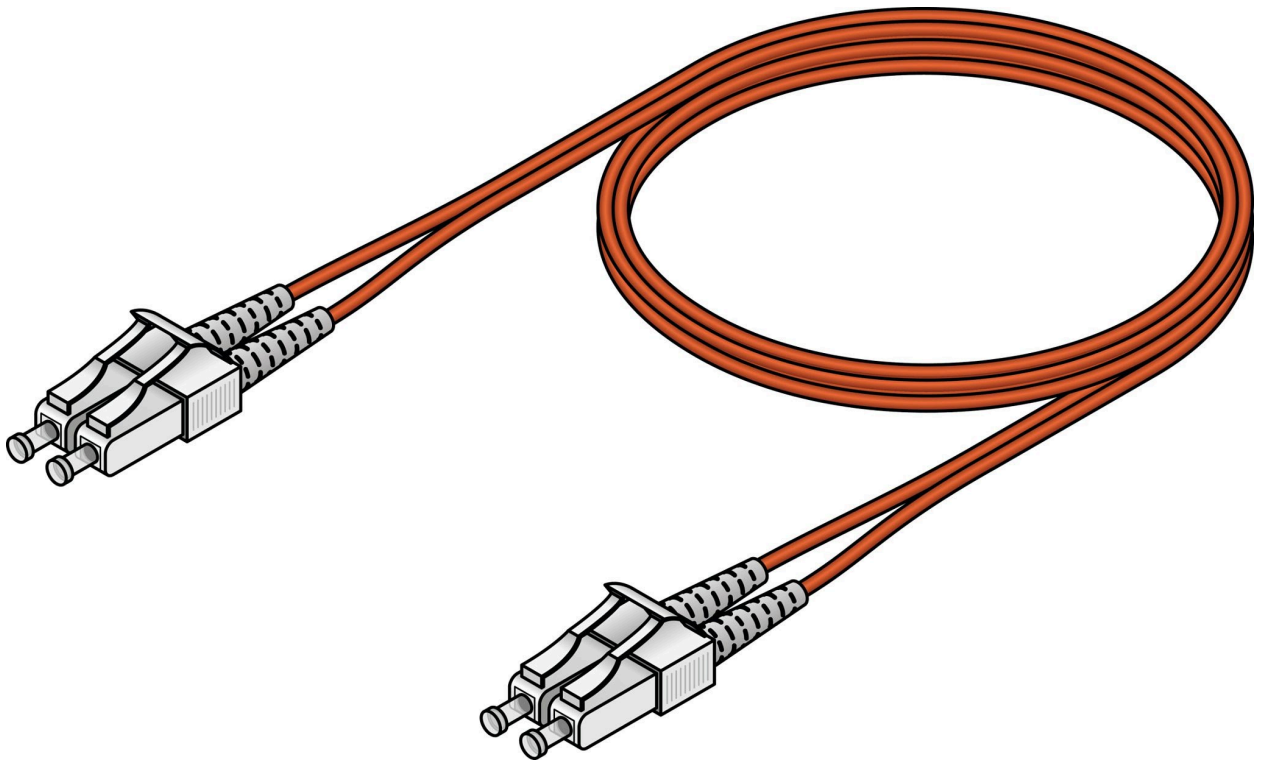
1. Ethernet Cable (Twisted Pair / Cat5e/Cat6)



Source: [Getty Images](#)

Common Use: Used inside homes and offices to physically connect PCs, consoles, and smart TVs to a router or switch.

2. Fiber Optic Cable



Source: [Getty Images](#)

Common Use: Used by internet service providers and data centers for high-speed, long-distance data transmission using light.

3. Coaxial Cable



Source: [Getty Images](#)

Common Use: Used to connect internet modems to wall outlets, cable TV boxes, and security camera systems.

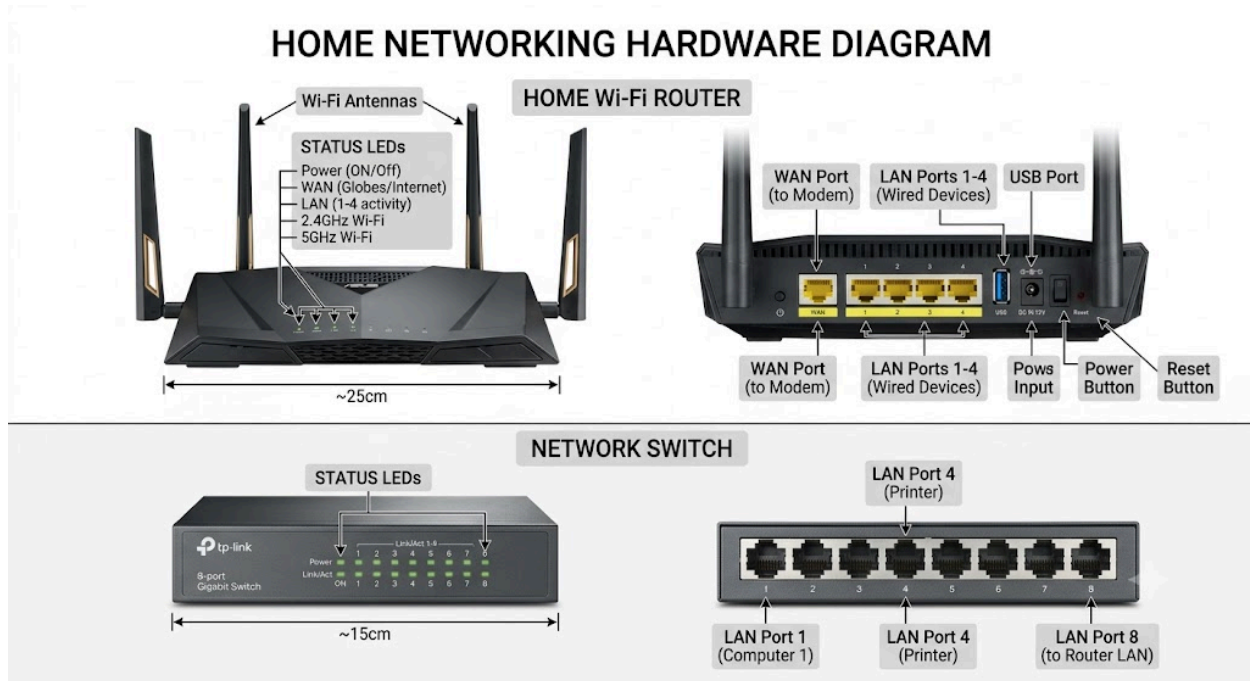
Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

ANS:

Feature	Router	Switch
Main Use	Connects different networks (LAN to Internet)	Connects devices inside the same network (LAN)
Data Handling	Uses IP addresses (Layer 3)	Uses MAC addresses (Layer 2)
Location	Network edge / Internet gateway	Center of a local network / Office server rack
Ports	Few ports (2–5) + Wi-Fi	High port density (8, 24, 48 wired ports)

DAY 2

Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

ANS: Network Types Connected: A **router** connects distinct networks together (e.g., your local home network to the public internet). A **switch** connects multiple devices within the *same* local network (LAN).

Addressing Used: A **router** inspects **IP addresses** (Layer 3) to route data between different networks. A **switch** inspects **MAC addresses** (Layer 2) to direct traffic to specific devices on the same network.

Traffic Routing Method: A **router** determines the best path to send data packets across complex networks. A **switch** forwards data frames directly from one port to another based on its internal MAC address table.

Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

ANS:



Router

- **Power Port:** Back panel; plugs into wall power.
- **Ethernet Ports:** Back panel; WAN connects to modem, LAN connects to local devices.
- **LED Indicators:** Front panel; shows power, internet link, and Wi-Fi activity.

Switch

- **Power Port:** Back/side panel; plugs into wall power.
- **Ethernet Ports:** Front panel; connects multiple local wired devices.
- **LED Indicators:** Front panel (or on ports); shows power, link connection, and port activity.

Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

ANS:

Setup:

Connect the **router** to the ISP modem line, then run a cable from the router to an **8+ port switch**, and plug all 8 gaming PCs into the switch.

Justification:

- **Router First:** Receives internet from the ISP, creates the local network, and manages traffic/firewall for IPL streaming.
- **Switch for PCs:** Expands wired ports and delivers low-latency, full-duplex connections to eliminate lag during multiplayer gaming.

DAY 3

Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

ANS:

Find IP: Run `ipconfig` in Command Prompt to find your **Default Gateway**.

Open Page: Type that IP (e.g., `192.168.1.1`) into your browser address bar.

Log In: Enter the admin credentials (usually printed on the router sticker).

Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

ANS:

Windows

- Open **Command Prompt** (press `Win + R`, type `cmd`, press Enter).
- Type `ipconfig` and press Enter.
- Look under **Wireless LAN adapter Wi-Fi** for your **IPv4 Address**, **Subnet Mask**, and **Default Gateway**.

Mac

- Open **System Settings > Wi-Fi > click Details** next to your connected network.
- Select **TCP/IP** to view your IP address, Subnet Mask, and Router (Default Gateway).

Android

- Go to **Settings > Network & internet (or Connections) > Wi-Fi**.
- Tap the gear icon next to your connected Wi-Fi > tap **View more / IP settings**.

iPhone / iPad

- Go to **Settings > Wi-Fi**.
- Tap the **(i)** info icon next to your connected Wi-Fi network.

Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

ANS:

Set Static IP: Go to Wi-Fi properties > IPv4 > Enter IP (192.168.1.50), Subnet (255.255.255.0), and Gateway (192.168.1.1).

Test Access: Open a browser and load a website to verify internet connectivity.

Revert to DHCP: Re-select **Obtain an IP address automatically** and save.

In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

ANS:

Locating the DHCP Range: Look under **LAN Settings** or **DHCP Server** in your router's admin panel (e.g., 192.168.1.100 to 192.168.1.250).

IP Conflict Impact: If two devices get the same IP, an **IP address conflict** occurs. Both devices will lose stable internet access, drop packets, or be disconnected completely because the router won't know where to send network traffic.

DAY 4

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

ANS:

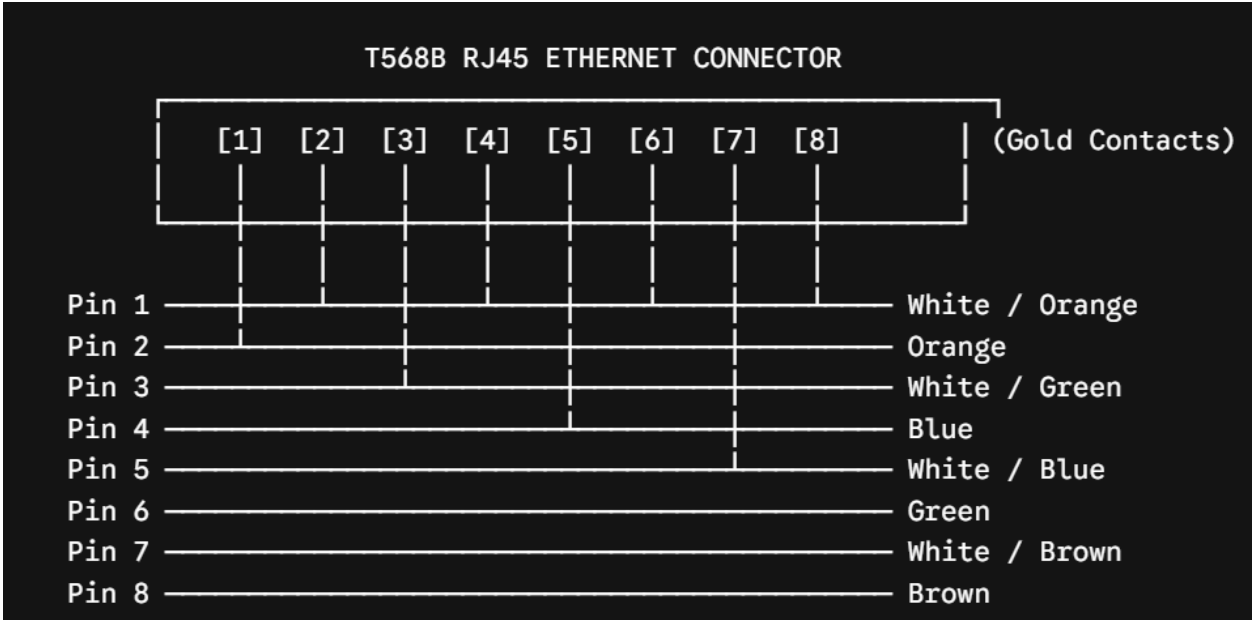
- **Transmission Medium:** Ethernet uses copper wires with electrical signals; fiber optic uses glass/plastic strands with light signals.
- **Speed & Distance:** Ethernet supports up to 10 Gbps over ~100 meters; fiber optic supports 100+ Gbps over tens of kilometers without signal loss.
- **Interference:** Ethernet is vulnerable to electromagnetic interference (EMI); fiber optic is completely immune to EMI.

Real-World Examples:

- **Ethernet:** Connecting a PC directly to a home Wi-Fi router.
- **Fiber Optic:** Linking buildings across a college campus backbone or connecting servers inside a data center.

Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

ANS:



Pin Number	Wire Color	Signal / Function
Pin 1	White / Orange	Transmit Data (+)
Pin 2	Orange	Transmit Data (-)
Pin 3	White / Green	Receive Data (+)
Pin 4	Blue	Unused / Power over Ethernet (PoE)
Pin 5	White / Blue	Unused / Power over Ethernet (PoE)

Pin 6	Green	Receive Data (-)
Pin 7	White / Brown	Unused / Power over Ethernet (PoE)
Pin 8	Brown	Unused / Power over Ethernet (PoE)

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

ANS:

Strip: Strip 3 cm of outer jacket and cut internal spline/threads.

Arrange: Untwist wires and align in T568B order (W/Orange, Orange, W/Green, Blue, W/Blue, Green, W/Brown, Brown).

Trim & Insert: Cut wires straight to ~1.2 cm and slide into the RJ45 connector (clip facing down) until they hit the top.

Crimp & Test: Squeeze firmly with the crimping tool, then test both ends with a cable tester.

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

ANS:

For a 10-PC LAN gaming café, **Ethernet cable (Cat6)** paired with an **RJ45 crimper** is the ideal choice. Cat6 Ethernet easily delivers up to 10 Gbps speeds across the short distances inside a single room while being significantly cheaper and easier to terminate than fiber. Fiber optic cables and specialized fiber cleavers are unnecessarily expensive for short LAN runs and offer no tangible performance benefit over copper for a local 10-PC network.

DAY 5

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use

'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

ANS:

Find Phone's IP: On phone, go to **Settings > Wi-Fi > tap network > note IP** (e.g., **192.168.1.15**).

Ping Phone: On laptop CMD, type **ping [Phone IP]** and press Enter.

Verify: Receiving **"Reply from..."** means communication works; **"Request timed out"** means AP isolation or firewall is blocking it.

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

ANS:

Windows: Run **tracert www.youtube.com** in Command Prompt.

Mac/Linux: Run **traceroute www.youtube.com** in Terminal.

Result: It typically takes **8–15 hops** (Hop 1 = your local router, final hop = Google/YouTube server).

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

ANS:

Command: Run **ping [Disconnected Device IP]** in Command Prompt.

Error Message: **"Request timed out"** (or **"Destination host unreachable"**).

Reason: The router cannot deliver packets because the target IP address is no longer active on the network.

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

ANS:

Run: Type **arp -a** in Command Prompt.

Match: Look at the **Internet Address** column and match each MAC address to your home devices:

IP Address	Device / Person
192.168.1.1	Router (Gateway)
192.168.1.15	My Laptop
192.168.1.20	Smartphone
192.168.1.50	Gaming PC / Smart TV

Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

ANS:

Project: 10-PC Gaming Café Network (Star Topology: Router → Switch → PCs)

Setup & Configuration

- **Cabling:** Terminated Cat6 cables using **T568B** standard (W-Org, Org, W-Grn, Blu, W-Blu, Grn, W-Brn, Brn). Verified pin continuity with a tester.
- **Hardware:** Connected ISP modem to router WAN. Connected router LAN to a 16-port Gigabit switch, feeding 10 gaming PCs.
- **IP Scheme:** Set router DHCP pool to 192.168.1.100–192.168.1.200 (Subnet 255.255.255.0, Gateway 192.168.1.1). Reserved static IP 192.168.1.50 for local server.

Diagnostics & Command Output

1. Local Ping Test (ping 192.168.1.1)

Plaintext

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

2. Path Analysis (**tracert** [www.youtube.com](<https://www.youtube.com>))

Plaintext

```
1  1 ms  1 ms  1 ms  192.168.1.1 (Local Gateway)
2  10 ms  9 ms  11 ms  10.240.0.1 (ISP Gateway)
3  22 ms  21 ms  20 ms  142.250.190.46 (YouTube Server)
```

Trace complete.

Results & Issues

- **Success:** All 10 PCs received dynamic IPs with <1ms local latency and 0% packet loss.
- **Issue 1 (AP Isolation):** Wireless devices timed out when pinging wired PCs. Fixed by disabling **AP Isolation** in router settings.
- **Issue 2 (Disconnection):** Unplugging a node returned **Destination host unreachable**, confirming offline status